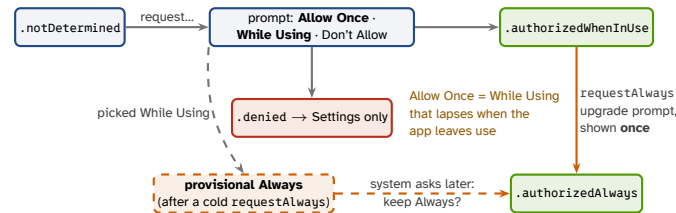


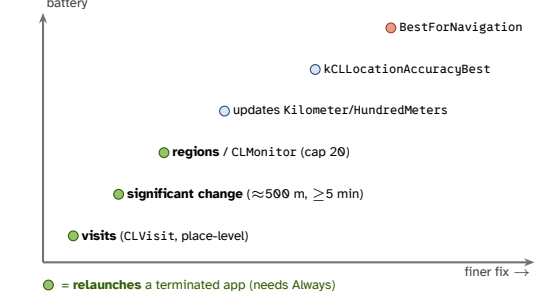
**In one line:** Core Location fuses GPS/Wi-Fi/cell into fixes under a permission the user can **shrink any time** (While Using/Always and Precise/Approximate); the right service is the **cheapest one that answers the question**. MapKit draws it: **annotations** = points (reused views), **overlays** = shapes (renderers).

### Authorization (iOS 13+)



orthogonal (iOS 14): `accuracyAuthorization`, `fullAccuracy` / `reducedAccuracy` (km-scale)  
→ `requestTemporaryFullAccuracy(withPurposeKey:)` — key in `NSLocationTemporaryUsage`..

### Which service? — power vs precision



## Core Location

- `CLLocationManager`: strong ref, created on main (callbacks use the creating thread's run loop). `locationManagerDidChangeAuthorization` (iOS 14) fires at creation too — read `authorizationStatus` there.
- `Info.plist` `NSLocationWhenInUseUsageDescription` (+ `...AlwaysAndWhenInUse...` for Always); missing → no prompt.
- `requestLocation()` = one fix, then stops (needs `didFailWithError`). `startUpdatingLocation()` = stream, tuned by `desiredAccuracy`, `distanceFilter`, `activityType`.
- **iOS 17:** `CLLocationUpdate.liveUpdates()` = `AsyncSequence`, no delegate; a held `CLBackgroundActivitySession` keeps it alive in the background. `CLMonitor` (actor) replaces `startMonitoring(for:)`: `add(condition, identifier:)`, then iterate events (`.satisfied/.unsatisfied`).
- **iOS 18:** `CLServiceSession(authorization: .whenInUse)` declares what a feature *needs*; Core Location prompts as appropriate. Iterating `liveUpdates/events` holds one implicitly (off with `NSLocationRequireExplicitServiceSession`). **Always** only takes effect while a `.always` session is held — started in the foreground.
- **Background:** mode location + `allowsBackgroundLocationUpdates = true`, started in the foreground.
- Quality: `horizontalAccuracy < 0` = invalid; first fix often cached (check timestamp). `CLGeocoder`: network, rate-limited.

## MapKit

- **Annotation** (`MKAnnotation` point) → `viewFor:` → **reused** `MKAnnotationView` (`register + dequeueReusable...`; `MKMarkerAnnotationView`). **Overlay** (`MKPolyline/Polygon/Circle/MKTileOverlay`) → `rendererFor:` → `MKOverlayRenderer`, scales with zoom.
- **Clustering** (iOS 11): shared `clusteringIdentifier` → `MKClusterAnnotation`; `displayPriority`.
- **Search:** `MKLocalSearchCompleter` (as you type) → `MKLocalSearch` → `[MKMapItem]`. **Directions:** `MKDirections.Request` → `calculate` → `MKRoute` (`polyline`, `expectedTravelTime`); or hand off: `openInMaps(launchOptions:)`.
- **SwiftUI Map (iOS 17):** content builder (`Marker`, `Annotation`, `MapPolyline`, `MapCircle`, `UserAnnotation`), `MapCameraPosition(.automatic, .region)`, `.userLocation(fallback:)`, `.onMapCameraChange`,

`.mapControls`. iOS 16 `Map(coordinateRegion:)` is deprecated.

### Example — iOS 17 async + SwiftUI Map

```
let bg = CLBackgroundActivitySession() // hold: bg updates
for try await u in CLLocationUpdate.liveUpdates() {
  guard let loc = u.location else { continue }
  track(loc) } // + u.isStationary
let m = await CLMonitor("stores") // actor, 20 max
await m.add(CLMonitor.CircularGeographicCondition(
  center: shop, radius: 150), identifier: "shop-7")
for try await e in await m.events {
  if case .satisfied = e.state { arrived(e.identifier) } }
Map(position: $camera, selection: $picked) {
  ForEach(shops) { s in
    Marker(s.name, coordinate: s.coord).tag(s.id) }
  MapPolyline(route.polyline).stroke(.blue, lineWidth: 4)
  UserAnnotation() }
```

## Traps

- ! Reading `authorizationStatus` right after `request...` — still `.notDetermined`; the answer comes in the delegate.
- ! No first prompt offers "Always". A cold `requestAlways` gives *provisional* Always; the upgrade prompt shows **once** — after that, and after `Don't Allow`, only Settings.
- ! **Blue pill:** mandatory for While-Using background use; off by default for Always (opt in via `showsBackgroundLocationIndicator`).
- ! `allowsBackgroundLocationUpdates` without the location mode crashes; `pausesLocationUpdatesAutomatically` silently stops a background tracker.
- ! **20 regions** per app: monitor the nearest 19 + one big region round the user whose *exit* re-plans. Geofences lag by minutes; radii under  $\approx 100$  m are unreliable.
- ! Returning a view for `MKUserLocation` hides the blue dot — return `nil`. `CLGeocoder` per keystroke is throttled.

### desiredAccuracy constants

|                                                  |                                 |                                       |
|--------------------------------------------------|---------------------------------|---------------------------------------|
| <code>...BestForNavigation</code>                | <code>...Best</code>            | highest (navigation) · best available |
| <code>...NearestTenMeters</code>                 | <code>...HundredMeters</code>   | 10 m · 100 m                          |
| <code>...Kilometer</code>                        | <code>...ThreeKilometers</code> | 1 km · 3 km                           |
| <code>kCLLocationAccuracyReduced</code> (iOS 14) |                                 | approximate, as if Precise off        |

## Remember

**Cheapest service that answers it** · permission can **shrink** any time · **points** = reused views, **shapes** = renderers.